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Nodera et al.

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(54) **TONER CONTAINER AND IMAGE FORMING APPARATUS**

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(52) **U.S. Cl.**
CPC ***G03G 15/0872*** (2013.01); ***G03G 15/0863*** (2013.01); *G03G 2215/0668* (2013.01); *G03G 2215/0675* (2013.01); *G03G 2215/0697* (2013.01)

(58) **Field of Classification Search**
CPC G03G 15/0863; G03G 15/0872
USPC 399/120, 252, 262
See application file for complete search history.

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(57) **ABSTRACT**

A toner container is attachable to and detachable from a body of an image forming apparatus and includes a substantially cylindrical container body and a substantially cylindrical cap. The container body conveys toner stored inside the container body. The cap has a through hole through which a toner conveying nozzle disposed in the body of the image forming apparatus is to be inserted into the container body. The through hole is in a central portion of the cap. The cap has a fitted portion to be fitted with a fitting portion disposed in the body of the image forming apparatus to position the cap non-rotatably to the body of the image forming apparatus. The fitted portion is disposed in each of an upper portion and a lower portion of the cap when the cap is attached to the body of the image forming apparatus.

17 Claims, 5 Drawing Sheets

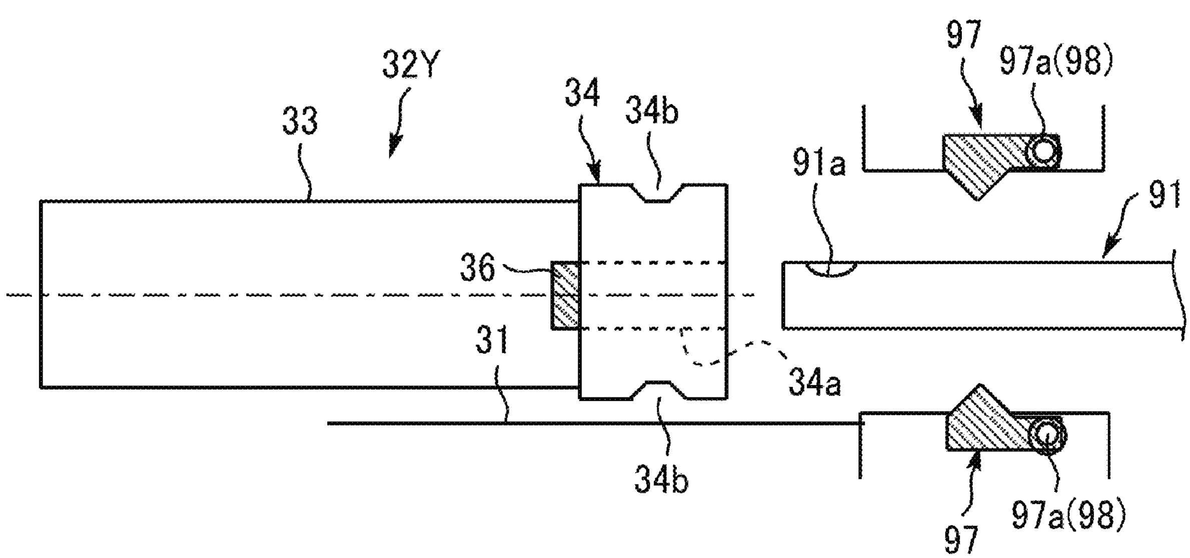

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FIG. 1

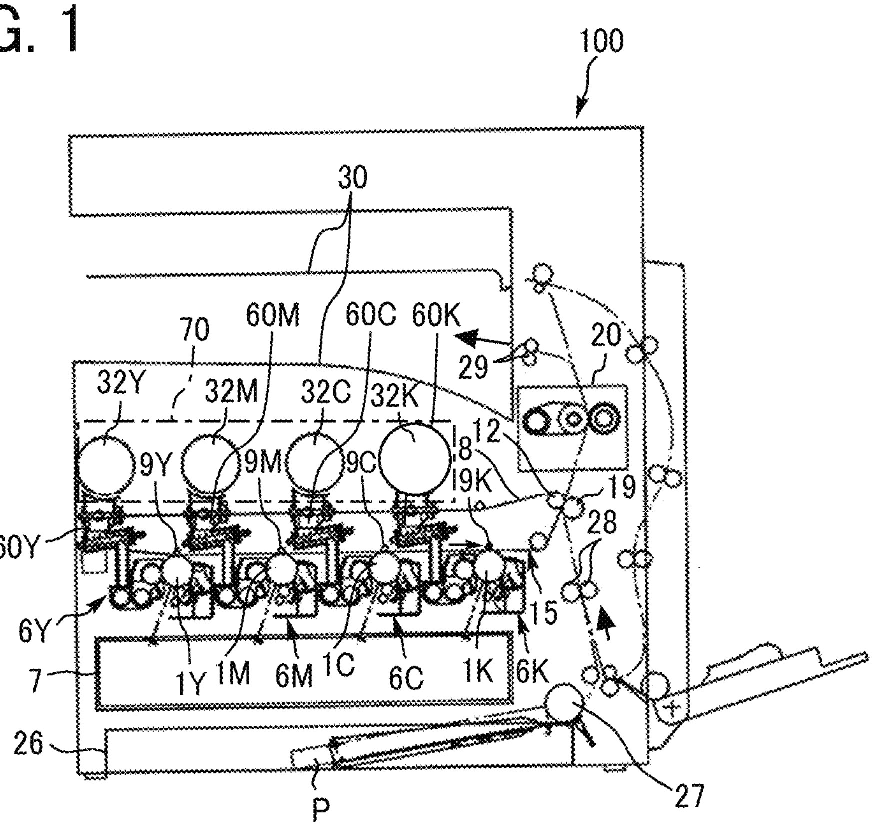

100
30
70
60M
60C
60K
20
29
32Y
32M
32C
32K
12
18
9Y
9M
9C
19K
19
28
60Y
6Y
15
7
1Y
1M
6M
1C
6C
1K
6K
26
P
27

FIG. 2

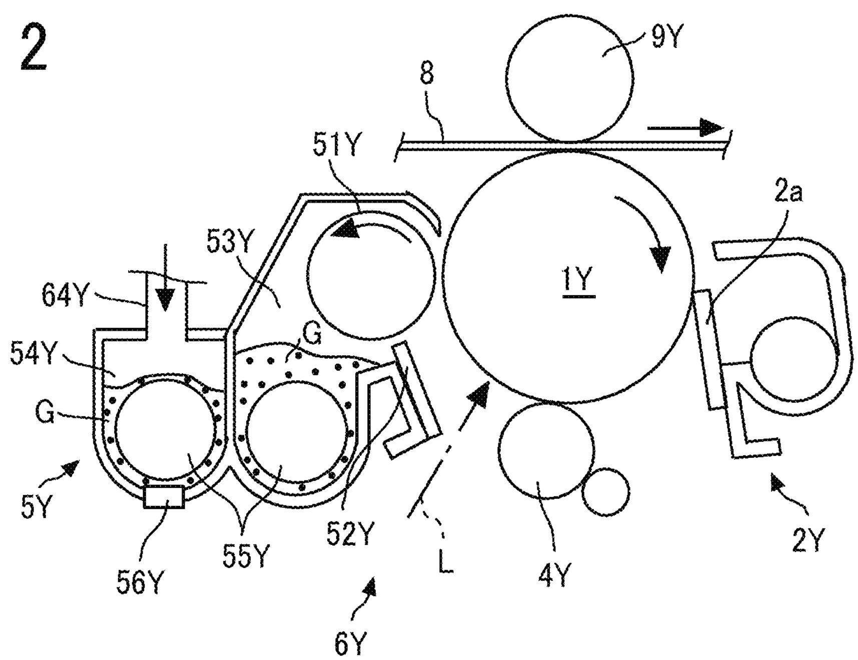
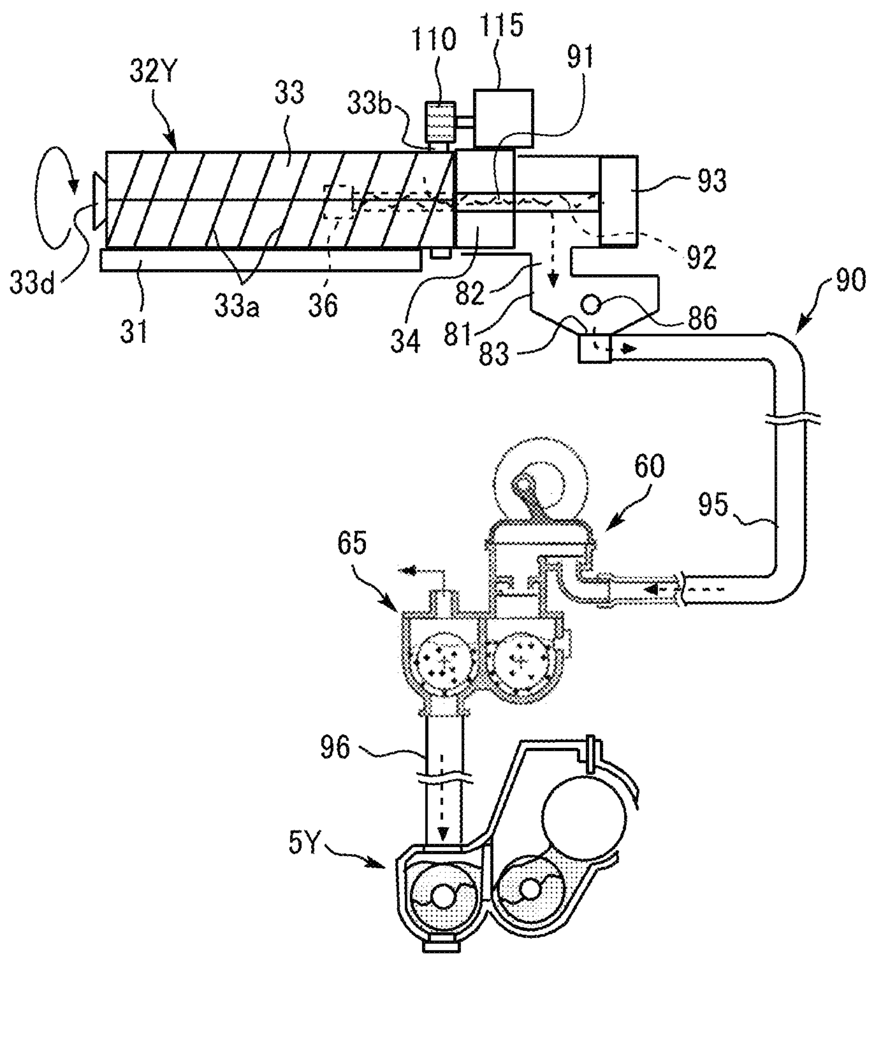

9Y
8
51Y
2a
53Y
1Y
64Y
G
54Y
G
5Y
56Y
55Y
52Y
L
4Y
2Y
6Y

FIG. 3



110
115
91
32Y
33
33b
93
33d
31
33a
36
34
82
81
83
92
86
90
60
95
65
96
5Y

FIG. 4A

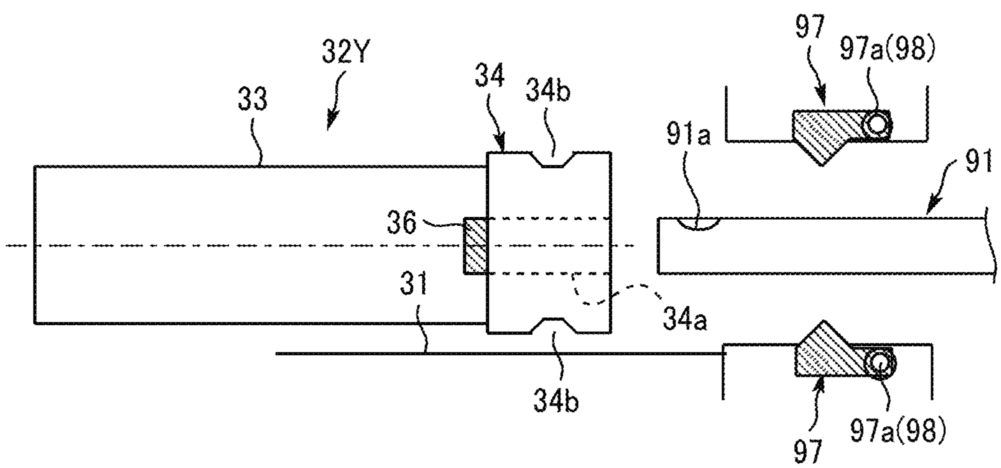

97
97a(98)
32Y
34
34b
33
91a
91
36
31
34a
34b
97a(98)
97

FIG. 4B

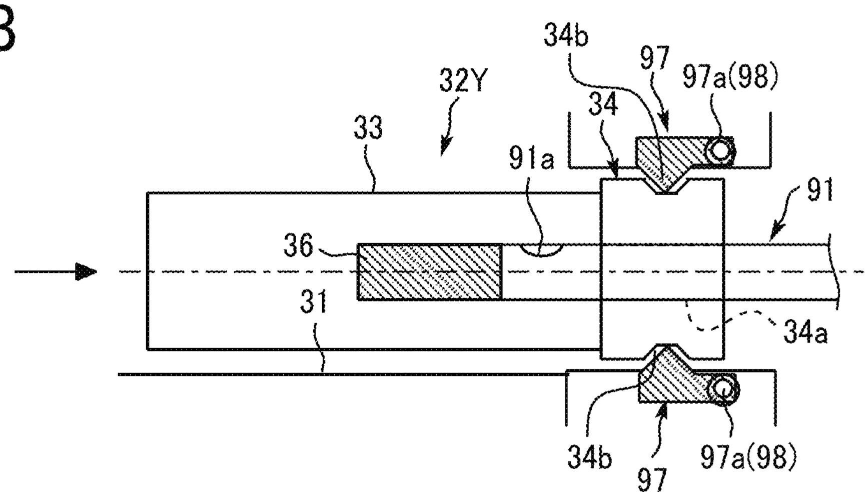

34b
97
97a(98)
32Y
34
33
91a
91
36
31
34a
34b
97
97a(98)

FIG. 5

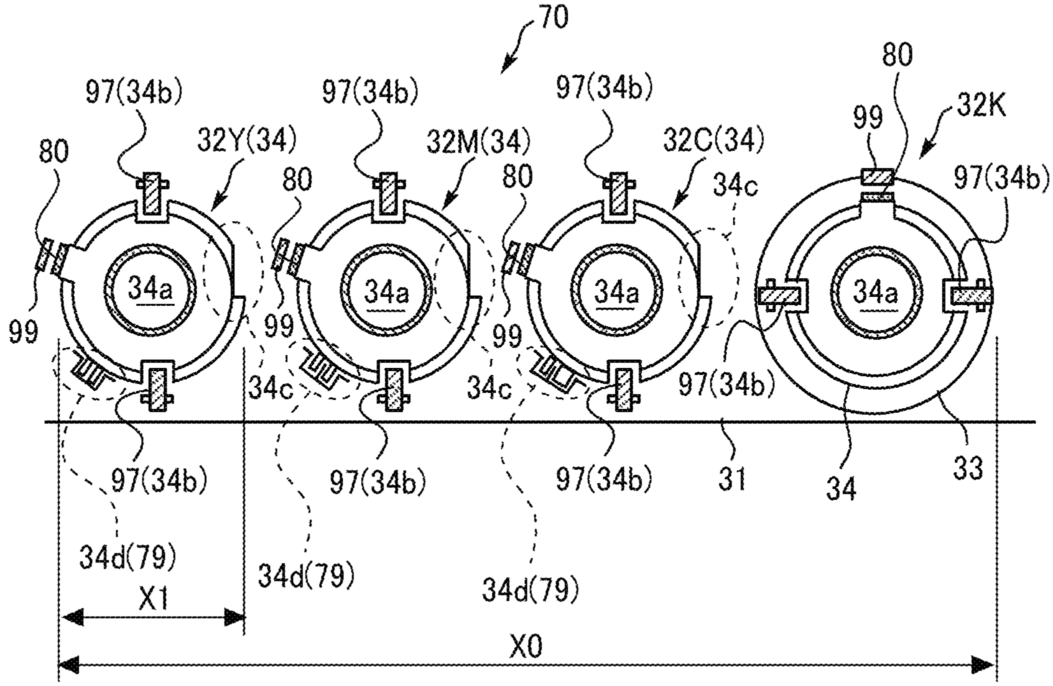


FIG. 6A

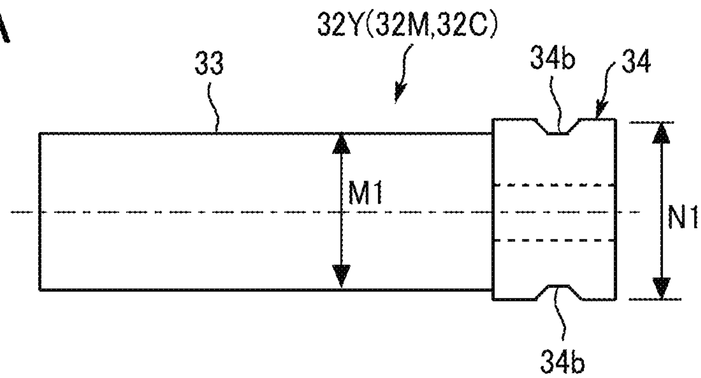
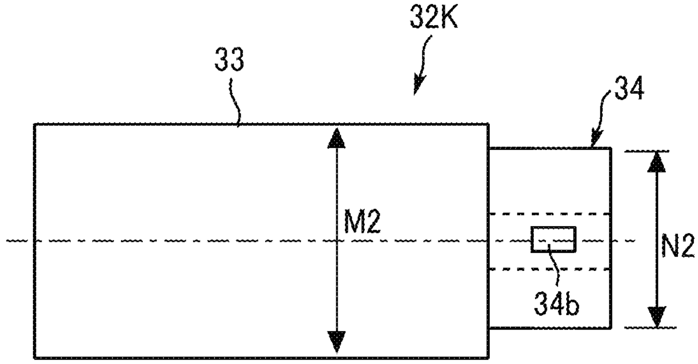


FIG. 6B



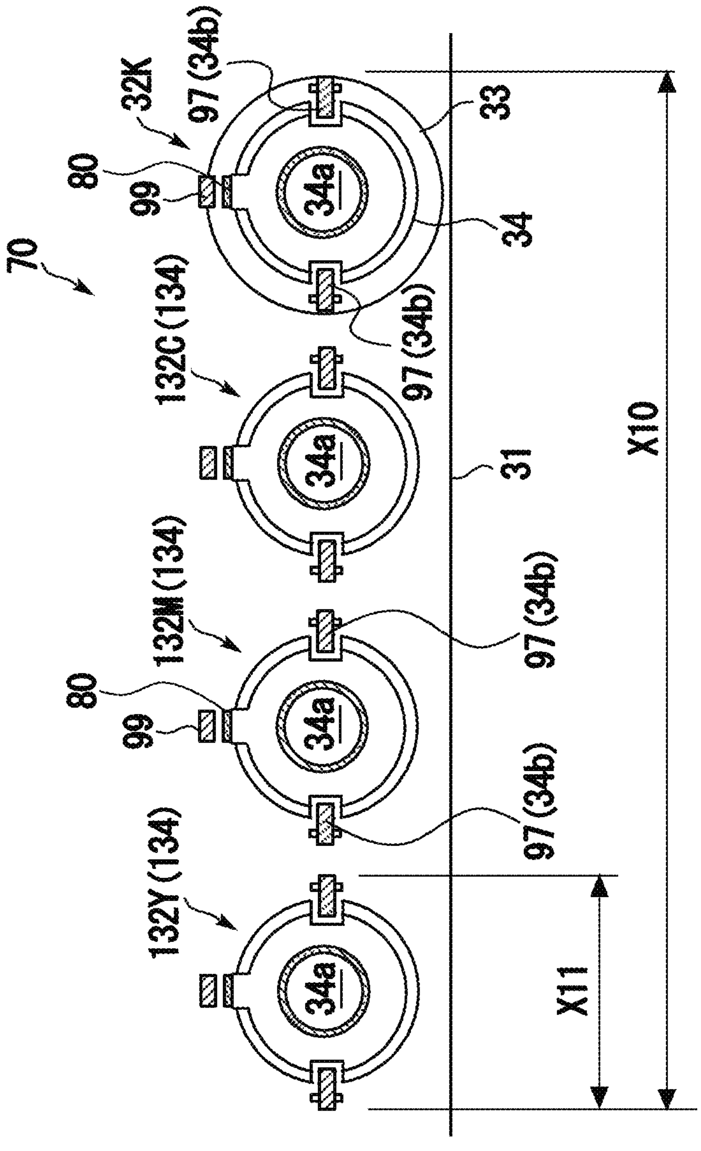

70
32K
80
99
97 (34b)
33
34
34a
97 (34b)
132C (134)
34a
31
X10
132M (134)
80
99
34a
97 (34b)
97 (34b)
132Y (134)
34a
X11

FIG. 7

TONER CONTAINER AND IMAGE FORMING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATION

This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2023-019009, filed on Feb. 10, 2023, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

Embodiments of the present disclosure relate to a toner container that stores toner therein, and an image forming apparatus such as a copier, a printer, a facsimile, or a multifunction peripheral thereof including the toner container.

Related Art

Image forming apparatuses such as copiers and printers are widely known in which a substantially cylindrical toner container attached to and detached from a body of an image forming apparatus is disposed.

SUMMARY

In an embodiment of the present disclosure, there is provided a toner container that is attachable to and detachable from a body of an image forming apparatus and includes a substantially cylindrical container body and a substantially cylindrical cap. The container body conveys toner stored inside the container body. The cap has a through hole through which a toner conveying nozzle disposed in the body of the image forming apparatus is to be inserted into the container body. The through hole is in a central portion of the cap. The cap has a fitted portion to be fitted with a fitting portion disposed in the body of the image forming apparatus to position the cap non-rotatably to the body of the image forming apparatus. The fitted portion is disposed in each of an upper portion and a lower portion of the cap when the cap is attached to the body of the image forming apparatus.

In another embodiment of the present disclosure, there is provided an image forming apparatus that includes a plurality of first toner containers and a second toner container. The plurality of first toner containers are arranged in a body of the image forming apparatus. Each of the plurality of first toner containers is attachable to and detachable from a body of the image forming apparatus and includes a substantially cylindrical container body and a substantially cylindrical cap. The container body conveys toner stored inside the container body. The cap has a through hole through which a toner conveying nozzle disposed in the body of the image forming apparatus is inserted into the container body. The through hole is in a central portion of the cap. The cap has a fitted portion to be fitted with a fitting portion disposed in the body of the image forming apparatus to position the cap non-rotatably to the body of the image forming apparatus. The fitted portion is disposed in each of an upper portion and a lower portion of the cap when the cap is attached to the body of the image forming apparatus. The second toner container is disposed adjacent to one of the plurality of first toner containers in the body of the image forming apparatus and includes a substantially cylindrical container body and a substantially cylindrical cap. The container body conveys toner stored inside the container body. The cap has a through hole through which a toner conveying nozzle disposed in the body of the image forming apparatus is inserted into the container body. The through hole is in a central portion of the cap. The cap has a fitted portion to be fitted with a fitting portion disposed in the body of the image forming apparatus to position the cap non-rotatably to the body of the image forming apparatus.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

FIG. **1** is a schematic view of an image forming apparatus according to an embodiment of the present disclosure;

FIG. **2** is a cross-sectional view of an image forming device of the image forming apparatus in FIG. **1**;

FIG. **3** is a diagram illustrating a toner container installed on a toner supply device, according to an embodiment of the present disclosure;

FIGS. **4**A and **4**B are side views of a toner container, illustrating operations of attaching the toner container to a body of an image forming apparatus, according to an embodiment of the present disclosure;

FIG. **5** is a front view of a plurality of toner containers attached to a body of an image forming apparatus according to an embodiment of the present disclosure;

FIG. **6**A is a side view of a toner container for primary color;

FIG. **6**B is a side view of a toner container for black (second toner container); and

FIG. **7** is a front view of a plurality of toner containers attached to a body of an image forming apparatus, as a comparative example.

The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. Like reference signs are assigned to like elements or components and descriptions of those elements or components may be simplified or omitted.

A configuration and operation of an image forming apparatus **100** is described below. As illustrated in FIG. **1**, four toner containers **32**Y, **32**M, **32**C, and **32**K corresponding to colors of yellow, magenta, cyan, and black, respectively, are detachably (i.e., replaceably) arranged in a toner container housing **70** (see FIG. **5**) above a body of the image forming apparatus **100**. These toner containers **32**Y, **32**M, **32**C, and **32**K are manually attached to and detached from the body of the image forming apparatus **100** by an operator such as a user. An intermediate transfer unit **15** is disposed below the toner container housing **70**. Image forming devices **6**Y, **6**M, **6**C, and **6**K corresponding to colors of yellow, magenta, cyan, and black, respectively, are arranged side by side to face an intermediate transfer belt **8** of the intermediate transfer unit **15**. Toner supply devices **90** are disposed below the toner containers **32**Y, **32**M, **32**C, and **32**K. Toner contained in the toner containers **32**Y, **32**M, **32**C, and **32**K is supplied (replenished) by the toner supply devices **90**, into developing devices of the image forming devices **6**Y, **6**M, **6**C, and **6**K.

With reference to FIG. **2**, the image forming device **6**Y for yellow includes a photoconductor drum **1**Y, a charging device **4**Y, a developing device **5**Y, a cleaning device **2**Y, and an electric-charge removing device disposed around the photoconductor drum **1**Y. Image forming processes (i.e., charging process, exposure process, development process, transfer process, cleaning process, and electric-charge removing process) are executed on the photoconductor drum **1**Y. Thus, a yellow toner image is formed on the surface of the photoconductor drum **1**Y.

The other three image forming devices **6**M, **6**C, and **6**K have substantially the same configuration as that of the image forming device **6**Y for yellow except for the color of toner used therein and the configuration of the toner container **32**K for black having a larger capacity, and form magenta, cyan, and black toner images. Only the image forming unit **6**Y is described below and descriptions of the other three image forming units **6**M, **6**C, and **6**K are appropriately omitted.

As illustrated in FIG. **2**, a drive motor drives to rotate the photoconductor drum **1**Y in a clockwise direction. The charging device **4**Y uniformly charges the surface of the photoconductor drum **1**Y (a charging process). When the surface of the photoconductor drum **1**Y reaches a position at which the surface of the photoconductor drum **1**Y is irradiated with a laser beam L emitted from an exposure device **7** (see FIG. **1**), the photoconductor drum **1**Y is scanned with the laser beam L at the position. Thus, an electrostatic latent image corresponding to yellow is formed on the photoconductor drum **1**Y (an exposure process).

When the surface of the photoconductor drum **1**Y reaches a position facing the developing device **5**Y, the electrostatic latent image is developed with toner into a yellow toner image (a development process). When the surface of the photoconductor drum **1**Y bearing the toner image reaches a position facing a primary transfer roller **9**Y via the intermediate transfer belt **8**, the toner image on the photoconductor drum **1**Y is transferred onto the intermediate transfer belt **8** (a primary transfer process). After the primary transfer process, a small amount of toner may remain untransferred on the photoconductor drum **1**Y.

When the surface of the photoconductor drum **1**Y reaches a position opposite the cleaning device **2**Y, a cleaning blade **2***a* of the cleaning device **2**Y mechanically collects the untransferred toner on the photoconductor drum **1**Y (a cleaning process). Finally, the surface of the photoconductor drum **1**Y reaches a position facing the electric-charge removing device, and residual potential on the photoconductor drum **1**Y is removed at this position (an electric-charge removing process). Thus, a series of image forming processes executed on the surface of the photoconductor drum **1**Y is completed.

The other image forming devices **6**M, **6**C, and **6**K perform the series of image forming processes described above in substantially the same manner as the image forming device **6**Y for yellow. In other words, the exposure device **7** disposed below the image forming devices **6**M, **6**C, and **6**K irradiates photoconductor drums **1**M, **1**C, and **1**K of the image forming devices **6**M, **6**C, and **6**K, respectively, with the laser beams L based on image data. Then, toner images formed on the photoconductor drums **1**Y, **1**M, **1**C, and **1**K through the development step are transferred and superimposed onto the intermediate transfer belt **8**. Thus, a color toner image is formed on the intermediate transfer belt **8**.

With reference to FIG. **1**, the intermediate transfer unit **15** includes the intermediate transfer belt **8**, the four primary transfer rollers **9**Y, **9**M, **9**C, and **9**K, a secondary-transfer counter roller **12**, multiple tension rollers, and an intermediate-transfer-belt cleaner. The intermediate transfer belt **8** is stretched around and supported by the multiple rollers and is rotated in the direction indicated by arrow illustrated in FIG. **1** as one of the multiple rollers that serves as a drive roller rotates (i.e., the secondary-transfer counter roller **12**).

The four primary transfer rollers **9**Y, **9**M, **9**C, and **9**K are pressed against the corresponding photoconductor drums **1**Y, **1**M, **1**C, and **1**K, respectively, via the intermediate transfer belt **8** to form primary transfer nips. A primary transfer bias opposite in polarity to toner is applied to the primary transfer rollers **9**Y, **9**M, **9**C, and **9**K. The intermediate transfer belt **8** travels in the direction (counterclockwise) indicated by an arrow in FIG. **1** and sequentially passes through the primary transfer nips of the four primary transfer rollers **9**Y, **9**M, **9**C, and **9**K. As a result, the single-color toner images on the photoconductor drums **1**Y, **1**M, **1**C, and **1**K, having the respective colors, are primarily transferred and superimposed onto the intermediate transfer belt **8**, thereby forming the multicolor toner image (a primary transfer process).

Subsequently, the intermediate transfer belt **8** onto which the toner images of the respective colors are transferred and superimposed reaches a position opposite a secondary transfer roller **19**. At this position, the intermediate transfer belt **8** is nipped between the secondary-transfer counter roller **12** and the secondary transfer roller **19** to form a secondary transfer nip. The toner images of four colors formed on the intermediate transfer belt **8** are transferred onto a sheet P such as a sheet of paper conveyed to the position of the secondary transfer nip (a secondary transfer process). At this time, the untransferred toner that has not been transferred onto the sheet P remains on the surface of the intermediate transfer belt **8**. The surface of the intermediate transfer belt **8** then reaches a position opposite the intermediate-transfer-belt cleaner. At the position, the intermediate-transfer-belt cleaner collects the untransferred toner from the surface of the intermediate transfer belt **8**. Thus, a series of transfer processes performed on the intermediate transfer belt **8** is completed.

The sheet P is conveyed from a sheet feeder **26** disposed in a lower portion of the apparatus body of the image forming apparatus **100** to the secondary transfer nip via a feed roller **27** and a registration roller pair **28**. Specifically, the sheet feeder **26** contains a stack of multiple sheets P such as sheets of paper stacked on one on another. As the feed roller **27** is rotated counterclockwise in FIG. **1**, the feed roller **27** feeds a top sheet P from the stack in the sheet feeder **26** to a roller nip between the registration roller pair **28**.

The sheet P conveyed to the registration roller pair **28** temporarily stops at a position of the roller nip between the rollers of the registration roller pair **28** that has stopped rotating. Rotation of the registration roller pair **28** is timed to convey the sheet P toward the secondary transfer nip such that the sheet P meets the color toner image on the intermediate transfer belt **8** at the secondary transfer nip. Thus, the desired color toner image is transferred onto the sheet P.

Subsequently, the sheet P, onto which the color toner image is transferred at the secondary transfer nip, is conveyed to a position of a fixing device **20**. Then, at this position, the color toner image transferred to the surface of the sheet P is fixed on the sheet P by heat and pressure of a fixing belt and a pressing roller. Subsequently, the sheet P is conveyed through the rollers of an output roller pair **29** and ejected to the outside of the image forming apparatus **100**. The sheets P ejected by the output roller pair **29** to the outside of the image forming apparatus **100** are sequentially stacked as output images on a stack tray **30**. Thus, a series of image forming processes performed by the image forming apparatus **100** is completed.

A detailed description is provided of a configuration and operation of the developing device **5**Y of the image forming device **6**Y with reference to FIG. **2**. The developing device **5**Y includes a developing roller **51**Y disposed opposite the photoconductor drum **1**Y, a doctor blade **52**Y disposed opposite the developing roller **51**Y, two conveying screws **55**Y disposed in developer housings **53**Y and **54**Y, and a toner concentration sensor **56**Y to detect concentration of toner in developer G. The developing roller **51**Y includes a magnet and a sleeve. The magnet is fixed inside the developing roller **51**Y. The sleeve rotates around the magnet. The developer housings **53**Y and **54**Y contain the two-component developer G including carrier (i.e., carrier particles) and toner (i.e., toner particles). The developer housing **54**Y communicates, via an opening on an upper side thereof, with a downward toner conveyance passage **64**Y.

The developing device **5**Y described above operates as follows. The sleeve of the developing roller **51**Y rotates in a direction indicated by an arrow in FIG. **2**. The developer G is carried on the developing roller **51**Y by a magnetic field generated by the magnet. As the sleeve rotates, the developer G moves along the outer circumferential surface of the developing roller **51**Y.

The developer G in the developing device **5**Y is adjusted so that the ratio of toner (toner concentration) in the developer G is within a specified range. More specifically, the toner supply device **90** (see FIG. **3**) supplies toner (as powder) from the toner container **32**Y to the developer housing **54**Y according to the toner consumption in the developing device **5**Y. The configuration and operation of the toner supply device are described in detail below.

The two conveying screws **55**Y mix and stir the developer G with the toner supplied to the developer housing **54**Y while circulating with the developer G in the two developer housings **53**Y and **54**Y. In this case, the developer G moves in the direction perpendicular to the surface of the plane on which FIG. **2** is illustrated. The toner in the developer G is electrically charged by friction with the carrier, so that the toner is attracted to the carrier. Both the toner and the carrier are borne on the developing roller **51**Y due to a magnetic force generated on the developing roller **51**Y.

The developer G borne on the developing roller **51**Y is conveyed in the direction indicated by arrow in FIG. **2** and reaches a position opposite the doctor blade **52**Y. At this position, the doctor blade **52**Y adjusts the amount of the developer G on the developing roller **51**Y to an appropriate amount. Thereafter, the developer G on the developing roller **51**Y is conveyed to a position opposite the photoconductor drum **1**Y (i.e., which is a development area). The toner is attracted to the electrostatic latent image formed on the photoconductor drum **1**Y by an electric field generated in the developing area. As the sleeve rotates, the developer G remaining on the developing roller **51**Y reaches an upper part of the developer housing **53**Y and separates from the developing roller **51**Y.

With reference to FIG. **3**, a description is given of the toner supply device **90** in detail below. The toner supply device **90** drives to rotate a container body **33** of the toner container **32**Y (serving as a powder container) disposed in a mount **31** of the toner container housing **70** (see FIG. **1**) in a specified direction (e.g., in the direction indicated by arrow in FIG. **3**), discharges the toner (serving as powder) contained in the toner container **32**Y to the outside of the toner container **32**Y, and guides the toner to the developing device **5**Y serving as a supplied portion via a sub-hopper **65**. The toner supply device **90** includes a toner supply path (i.e., a toner conveyance path). In FIG. **3**, the arrangement direction of the toner container **32**Y, the toner supply device **90**, and the developing device **5**Y is changed to facilitate understanding. Actually, the longitudinal axes of the toner container **32**Y and a part of the toner supply device **90** are perpendicular to the plane on which FIG. **3** is illustrated (see FIG. **1**). In addition, the orientation and arrangement of conveying tubes **95** and **96** are also simplified and illustrated.

The toner supply devices **90** supply the color toners contained in the toner containers **32**Y, **32**M, **32**C, and **32**K installed in the mount **31** in the body of the image forming apparatus **100** (the toner container housing **70**) to the corresponding developing devices **5**Y, **5**M, **5**C, and **5**K, respectively. The amount of toner supplied to each developing device **5** is determined based on the amount of toner consumed in the corresponding developing device **5**. The four toner supply devices **90** have substantially the same configuration except the color of the toner used in the image forming processes. Specifically, with reference to FIGS. **3**, **4**A, and **4**B, when the toner container **32**Y is attached to the mount **31** (the toner container housing **70**) of the body of the image forming apparatus **100**, a toner conveying nozzle **91** (as the nozzle) of the body of the image forming apparatus **100** pushes and moves a shutter **36** of the toner container **32**Y. As a result, the toner conveying nozzle **91** is inserted into the toner container **32**Y (the container body **33**) via a through hole **34***a*. Accordingly, the toner contained in the toner container **32**Y can be discharged through the toner conveying nozzle **91**. The toner container **32**Y includes a grip **33***d* at a bottom portion (on the left side in FIG. **3**) of the toner container **32**Y so that a user easily handles and installs the toner container **32**Y to the mount **31**. The grip **33***d* has an outer diameter smaller than the outer diameter of the container body **33**. The user grips the grip **33***d* to install the toner container **32**Y in the mount **31** and take out the toner container **32**Y from the mount **31**.

With reference to FIG. **3**, the toner container **32**Y includes the substantially cylindrical container body **33** having a spiral groove **33***a* extending in the longitudinal direction (the left and right direction in FIG. **3**, and the axial direction of the container body **33**). Specifically, the spiral groove **33***a* is formed from an outer circumferential surface toward an inner circumferential surface of the container body **33** so that a rotation of the container body **33** convey the toner in the container body **33** from the left to the right in FIG. **3**. The toner conveyed from the left to the right in FIG. **3** inside the container body **33** is discharged to the outside of the toner container **32**Y through the toner conveying nozzle **91**. The toner container **32**Y includes a gear **33***b* meshing with a drive gear **110** of the body of the image forming apparatus **100**. The gear **33***b* is disposed on the outer circumferential surface of the container body **33** on a head side (on the right side of the container body **33** in FIG. **3**). When the toner container **32**Y is installed to the mount **31**, the gear **33***b* of the container body **33** meshes with the drive gear **110** of the body of the image forming apparatus **100**. As a drive motor **115** is driven, the driving force (rotational drive) is transmitted from the drive gear **110** to the gear **33***b*, thus driving the container body **33** to rotate. In other words, the drive motor **115** and the drive gear **110** function as a drive mechanism to drive to rotate the container body **33**. A description is given of the toner container **32**Y in further detail below.

With reference to FIG. **3**, a conveying screw **92** is disposed inside the toner conveying nozzle **91**. As a motor **93** drives to rotate the conveying screw **92**, the conveying screw **92** conveys the toner flowing into the toner conveying nozzle **91** from an opening **91***a* (serving as an inlet port) in the toner container **32**Y from the left to the right in FIG. **3**. Thus, the toner is discharged through an outlet port of the toner conveying nozzle **91** to a hopper **81**. The hopper **81** is disposed below the outlet port of the toner conveying nozzle **91** via a dropping path **82**. The toner stored in the hopper **81** is conveyed to the developing device **5**Y downstream from the hopper **81** by a conveyor. The conveyance configuration by the conveyor in FIG. **3** is described below. A suction port **83** is disposed in the bottom portion of the hopper **81** and coupled to one end of the conveying tube **95** as a tube. The conveying tube **95** is made of a flexible rubber material with low affinity for toner, and the other end of the conveying tube **95** is coupled to a developer pump **60** (serving as a diaphragm pump). The developer pump **60** is coupled to the developing device **5**Y via the sub-hopper **65** and the conveying tube **96**. With such a configuration of the toner supply device **90**, the drive motor **115** as the drive mechanism rotates the container body **33** of the toner container **32**Y to discharge the toner stored in the toner container **32**Y to the outside of the toner container **32**Y through the toner conveying nozzle **91**. The toner discharged from the toner container **32**Y falls through the dropping path **82** and is stored in the hopper **81**. The developer pump **60** operates to suck the toner stored in the hopper **81** together with air from the suction port **83** and convey the toner from the developer pump **60** to the sub-hopper **65** through the conveying tube **95**. The toner conveyed to and stored in the sub-hopper **65** is appropriately supplied into the developing device **5**Y via the conveying tube **96**. In other words, the toner in the toner container **32**Y is conveyed in the direction indicated by dashed arrows in FIG. **3**. The conveyor is not limited to the above-described configuration, and for example, the toner stored in the hopper **81** may be conveyed directly to the developing device **5**Y by a screw disposed in the hopper **81**.

A toner sensor **86** is disposed near the suction port **83** and indirectly detects a state in which the toner contained in the toner container **32**Y is depleted (i.e., toner end state) or a state in which the toner contained in the toner container **32**Y is nearly depleted (i.e., toner near end state). The toner is discharged from the toner container **32**Y based on the detection result of the toner sensor **86**. Specifically, for example, a piezoelectric sensor or a light transmission sensor can be used as the toner sensor **86**. A height of the detection surface of the toner sensor **86** is set so that the amount of toner (deposition height) deposited above the suction port **83** is a target value. A drive timing and a drive duration of the drive motor **115** are controlled to rotationally drive the toner container **32**Y (container body **33**) based on the detection result of the toner sensor **86**. Specifically, when the toner sensor **86** detects that toner is not deposited on a detection position of the toner sensor **86**, the drive motor **115** is driven for a specified time. When the toner sensor **86** detects that toner is present on the detection position, the drive motor **115** stops. If the toner sensor **86** continuously detects that toner does not exist at the detection position even when the above-described control is performed repeatedly, a controller of the image forming apparatus determines that the toner contained in the toner container **32**Y is depleted (i.e., toner end state) or that the toner contained in the toner container **32**Y is nearly depleted (i.e., toner near end state).

A description is given of a configuration and an operation of the toner container **32**Y (**32**M, **32**C, or **32**K) according to the present embodiment below. As described above, in the present embodiment, the toner containers **32**Y, **32**M, **32**C, and **32**K for four colors are installed to be attachable to and detachable from the body of the image forming apparatus **100** (the toner container housing **70**). The three toner containers **32**Y, **32**M, and **32**C (serving as first toner containers) for primary colors (yellow, magenta, and cyan) are formed in substantially the same shape except for an incompatible-shape portion **34***d* (which can be installed by the secondary processing) described below, and are commonized. As illustrated in FIG. **5**, the three toner containers **32**Y, **32**M, and **32**C for primary colors are arranged in parallel in a substantially horizontal direction (lateral direction) in the toner container housing **70** (the body of the image forming apparatus **100**). On the other hand, as illustrated in FIGS. **5**, **6**A and **6**B, the toner container **32**K for black (serving as a second toner container) is not commonized with the toner containers **32**Y, **32**M, and **32**C for primary colors, and is increased in capacity due to frequent use of printing using black. As illustrated in FIG. **5**, the black toner container **32**K (second toner container) is installed in the toner container housing **70** (the body of the image forming apparatus **100**) to be adjacent to one (the cyan toner container **32**C) of the plurality of toner containers **32**Y, **32**M, and **32**C for primary colors in a substantially horizontal direction (lateral direction).

Specifically, as illustrated in FIG. **6**A, the toner containers **32**Y, **32**M, and **32**C (first toner containers) for primary colors are formed such that the outer diameter M**1** of the container body **33** thereof and the outer diameter N**1** of a cap **34** thereof substantially match each other (M**1**≈N**1**). As illustrated in FIG. **6**B, in the present embodiment, the outer diameter M**2** of the container body **33** of the toner container **32**K (second toner container) for black is larger than the outer diameter N**2** of the cap **34** of the toner container **32**K (second toner container) for black (M**2**>N**2**). As illustrated in FIGS. **6**A and **6**B, the outer diameter M**1** of the container body **33** of any of the toner containers **32**Y, **32**M, and **32**C (first toner containers) for primary colors is smaller than the outer diameter M**2** of the container body **33** of the black toner container **32**K (second toner container) (M**1**>M**2**), and only the black toner container **32**K is increased in capacity. The outer diameter N**2** of the cap **34** of the toner container **32**K (second toner container) for black is substantially equal to the outer diameter N**1** of the cap **34** of the toner containers **32**Y, **32**M, and **32**C (first toner container) for primary colors (N**1**≈N**2**). In the specification of the present application, the above-described "outer diameter" of the container body **33** or the cap **34** is defined as an outer diameter of the outer circumferential surface in an imaginary shape that is substantially a cylinder, even if the container body **33** or the cap **34** is not a complete cylinder and has some irregularities on the outer circumferential surface.

More specifically, each of the three toner containers **32**Y, **32**M, and **32**C for primary colors and the toner container **32**K for black (second toner containers) includes a substantially cylindrical container body **33** and a substantially cylindrical (cap-shaped) cap **34**. The container body **33** conveys toner stored therein toward the cap **34**. With reference to FIG. **4**, the through hole **34***a* for inserting the toner conveying nozzle **91** disposed in the body of the image forming apparatus **100** (the toner container housing **70**) into the container body **33** is formed in a central portion (a portion corresponding to the rotation center of the container body **33**) of the cap **34**. In conjunction with the attachment of the toner container **32**Y, **32**M, **32**C, or **32**K to the body of the image forming apparatus **100**, in the cap **34**, a lever **97** serving as a fitting portion disposed in the body of the image forming apparatus **100** (the mount **31** of the toner container housing **70**) is fitted with a hole portion **34***b* serving as a fitted portion and is non-rotatably positioned relative to the body of the image forming apparatus **100** (the mount **31**).

As described above, in any of the toner containers **32**Y, **32**M, **32**C, and **32**K according to the present embodiment, toner is discharged to the outside of the toner container **32**Y, **32**M, **32**C, or **32**K by the toner conveying nozzle **91** inserted in the container body **33** via the through hole **34***a* formed in the central portion of the cap **34**. Thus, the failure that the lower portion of the cap **34** is contaminated with toner is less likely to occur than in the case where toner is discharged to the outside of the toner containers **32**Y, **32**M, **32**C, and **32**K from a toner ejection port formed at the bottom of the cap **34**.

With reference to FIGS. **4**A to **6**B, in the present embodiment, the hole portions **34***b* serving as the fitted portions are formed in the shape of grooves (or in the shape of through holes) at opposing positions (positions 180 degrees out of phase) on the outer circumferential surface of the cap **34**. The levers **97** serving as the fitting portions are disposed in the toner container housing **70** at positions corresponding to the two hole portions **34***b* in the cap **34** and are biased by torsion springs **98** serving as biasing members in a direction to fit the hole portions **34***b*. The lever **97** is disposed to be rotatable around a shaft **97***a*, and a winding portion of the torsion spring **98** (two leg portions are hooked to enable the above-described biasing) is inserted to the shaft **97***a*. The toner container housing **70** is provided with the lever **97**, a projection for restricting the rotation of the lever **97** in a biasing direction, and a housing case for holding, for example, a reading device **99** and a mount-side incompatible-shape portion **79** to be described below. As illustrated in FIG. **4**A, when the toner container **32**Y (**32**M, **32**C, or **32**K) is pushed in by a user toward the right side in FIG. **4**A (the rear side in the attachment direction of the toner container housing **70**), the cap **34** eventually pushes the lever **97** in a direction against the biasing direction. As illustrated in FIG. **4**B, when the hole portion **2***b* of the cap **34** has reached the position of the lever **97**, the lever **97** is fitted with the hole portion **2***b*, and the position of the cap **34** in the toner container housing **70** is determined (the cap **34** is held so as not to rotate in conjunction with the rotation of the container body **33**). The fitted portion and the fitting portion described above are not limited to those of the present embodiment, and various types of configurations can be used within the scope of the technical idea of the present disclosure.

With reference to FIG. **5** (and FIGS. **4**A, **4**B, **6**A, and **6**B), in the present embodiment, the hole portions **34***b* as the fitted portions of each of the three toner containers **32**Y, **32**M, and **32**C for primary colors are disposed in an upper portion and a lower portion of the cap **34** attached to the body of the image forming apparatus **100** (toner container housing **70**). The levers **97** as the fitting portions of the body of the image forming apparatus **100** (the toner container housing **70**) are disposed above and below the cap **34** so as to sandwich the cap **34**, in alignment with the two hole portions **34***b* of the cap **34** of each of the toner containers **32**Y, **32**M, and **32**C.

Thus, any of the three toner containers **32**Y, **32**M, and **32**C for primary colors is provided with the hole portions **34***b* (fitted portion) in the upper portion and lower portion of the cap **34**, and the levers **97** (fitting portions) are disposed at the positions corresponding to the hole portions **34***b* in the body of the image forming apparatus **100**. The lateral space X**1** for installing each of the toner containers **32**Y, **32**M, and **32**C (caps **34**) in the body of the image forming apparatus **100** (toner container housing **70**) can be set smaller than the lateral space X**11** for installing each of toner containers **132**Y, **132**M, and **132**C (caps **134**) in the case where hole portions **34***b* (fitted portions) are disposed on both sides of the cap **134** as in the toner containers **132**Y, **132**M, and **132**C illustrated as the comparative example in FIG. **7**. In other words, X**1**<X**11** is satisfied with reference to and comparison with FIGS. **5** and **7**. In particular, when the three toner containers **32**Y, **32**M, and **32**C (the caps **34**) are arranged in parallel in the lateral direction in the body of the image forming apparatus **100** (the toner container housing **70**), the difference in the three lateral spaces turn to be large. In other words, with reference to FIGS. **5** and **7**, the space (3×X**1**) for installing the three toner containers **32**Y, **32**M, and **32**C is significantly smaller than the space (3×X**11**) of the comparative example of FIG. **7** (3×X**1**<<3×X**11**). Unlike the toner container **32**K for black, the toner containers **32**Y, **32**M, and **32**C for primary colors have almost no difference of outer diameters between the container body **33** and the cap **34**. If the lever **97** (and a portion of the housing case that holds the lever **97**) of the body of the image forming apparatus **100** is placed in a space between the caps **34** adjacent to each other, the overall space in the lateral direction turns to be large. However, when the lever **97** of the body of the image forming apparatus **100** is not disposed in the space between the caps **34** adjacent to each other (e.g., the hole portions **34***b* are disposed at the upper portion and the lower portion in the cap **34**), the overall space in the lateral direction is reduced.

In contrast, the toner container **32**K for black (which is the second toner container attached to the body of the image forming apparatus **100**) is provided with the hole portions **34***b* (serving as the fitted portions) on the right side and the left side of the cap **34**, and a lever **97** (serving as the fitting portion) is disposed at the position corresponding to the hole portions **34***b* in the body of the image forming apparatus **100**. In the toner container **32**K (second toner container) for black, the outer diameter M**2** of the container body **33** is larger than the outer diameter N**2** of the cap **34**. As illustrated in FIG. **5**, when viewed from the cap **34** as the front side, the lateral space for arranging the levers **97** on both sides of the body of the image forming apparatus **100** causes such an arrangement to be adopted. As a whole, the lateral space X**0** for installing the four toner containers **32**Y, **32**M, **32**C, and **32**K in the image forming apparatus **100** illustrated in FIG. **5** is considerably smaller than the lateral space X**10** for installing the four toner containers **132**Y, **132**M, **132**C, and **132**K in the image forming apparatus **100** illustrated as the comparative example in FIG. **7** (X**0**<<X**10**). Accordingly, the image forming apparatus **100** (the toner container hous-

ing **70**) according to the present embodiment is reduced in size in the lateral direction as compared to that illustrated in FIG. **7**.

In the present embodiment, each of the caps **34** of the plurality of toner containers **32**Y, **32**M, and **32**C for primary colors is provided with an identification (ID) chip **80** that can communicate with the reading device **99** of the body of the image forming apparatus **100** in a side surface (a left side surface) farther from the toner container **32**K (second toner container) for black. This is to avoid interference of the ID chip **80** with the lever **97** (fitting portion) disposed in the body of the image forming apparatus **100**. The hole portions **34***b* (fitted portions) are disposed in the upper and lower portions of the cap **34** of each of the plurality of toner containers **32**Y, **32**M, and **32**C for primary colors. For the same reason, the cap **34** of the toner container **32**K (second toner container) for black is provided with the ID chip **80** at its top, which can communicate with the reading device **99** of the body of the image forming apparatus **100**. Information exchanged between the reading device **99** of the body of the image forming apparatus **100** and the ID chips **80** in the toner containers **32**Y, **32**M, **32**C, and **32**K includes information of the toner in the toner containers (e.g., color, capacity, and manufacturing lot) and information on the presence or absence of recycling.

With reference to FIG. **5**, in the present embodiment, the cap **34** of each of the plurality of toner containers **32**Y, **32**M, and **32**C for primary colors is provided with a cutout portion **34***c*, which is recessed as compared with the other portions of the circumferential surface of the cap, on a side surface (right side surface) closer to the toner container **32**K (second toner container) for black. The cutout portion **34***c* is recessed as compared with the other portions of the circumferential surface of the cap. Forming the cutout portion **34***c* in the outer circumferential portion of the cap **34** in this manner can prevent interference of the cap **34** with the ID chip **80** or the reading device **99** of the adjacent toner container.

With reference to FIG. **5**, in the present embodiment, the cap **34** of each of the plurality of toner containers **32**Y, **32**M, and **32**C for primary colors are provided with the incompatible-shape portion **34***d* for distinguishing the cap **34** from the caps **34** of the other two of the plurality of toner containers **32**Y, **32**M, and **32**C. In the body of the image forming apparatus **100** (toner container housing **70**), the mount-side incompatible-shape portion **79** that can fit the incompatible-shape portion **34***d* is disposed in accordance with the shape of the incompatible-shape portion **34***d*. Accordingly, even if a user attempts to install any one of the toner containers **32**Y, **32**M, and **32**C for primary colors to the mount **31** for another color, the incompatible-shape portion **34***d* interferes with the mount-side incompatible-shape portion **79** and prevents an installation error. Thus, color mixture due to the installation error can be avoided. The toner container **32**K for black has a larger capacity than any of the toner containers **32**Y, **32**M, and **32**C for primary colors, and the toner container **32**K as a whole is incompatible with any of the toner containers **32**Y, **32**M, and **32**C for primary colors.

As described above, the toner containers **32**Y, **32**M, and **32**C for primary colors according to the present embodiment are toner containers that are attachable to and detachable from the body of the image forming apparatus **100**. Each of the toner containers **32**Y, **32**M, and **32**C for primary colors is provided with the substantially cylindrical container body **33** that conveys toner stored therein toward the cap **34**. Each of the toner containers **32**Y, **32**M, and **32**C for primary colors is provided with the substantially cylindrical cap **34** that has the through hole **34***a* at a central portion thereof for inserting the toner conveying nozzle **91** disposed in the body of the image forming apparatus **100** into the container body **33**. When the levers **97** (fitting portions) disposed in the body of the image forming apparatus **100** are fitted with the hole portions **34***b*, the substantially cylindrical cap **34** is non-rotatably positioned in the body of the image forming apparatus **100**. The hole portion **34***b* is arranged in each of the upper and lower portions of the cap **34** attached to the body of the image forming apparatus **100**. As a result, the lateral space for installing the toner containers **32**Y, **32**M, and **32**C is less likely to increase in the body of the image forming apparatus **100**.

In the present embodiment, only toner is stored in the toner containers **32**Y, **32**M, **32**C, and **32**K. However, in some embodiments, toner containers may contain two-component developer including toner and carrier to be used in an image forming apparatus in which the two-component developer is appropriately supplied to developing devices. Even in such a case, similar effects to the above-described embodiments are also attained.

In the present embodiment, at least one or all of the image forming units **6**Y, **6**M, **6**C, and **6**K may be integrated as a process cartridge. Even in such a case, similar effects to the above-described embodiments are also attained. The term "process cartridge" used in the present disclosure defines a unit that includes an image bearer united with at least one of a charging device to charge the image bearer, a developing device to develop a latent image on the image bearer, and a cleaning device to clean the image bearer, and that is removably attached to a body of an image forming apparatus.

Embodiments of the present disclosure are not limited to the above-described embodiments and it is apparent that the above-described embodiments can be appropriately modified within the scope of the technical idea of the present disclosure in addition to what is suggested in the above-described embodiments. The number, position, shape, and so forth of components are not limited to those of the present embodiments and modifications, and may be the number, position, shape, and so forth that are suitable for implementing the present disclosure.

Aspects of the present disclosure may be, for example, combinations of first to ninth aspects as follows.

First Aspect

A toner container (e.g., the toner container **32**Y) is attachable to and detachable from a body of an image forming apparatus (e.g., the image forming apparatus **100**). The toner container includes a substantially cylindrical container body (e.g., the container body **33**) and a substantially cylindrical cap (e.g., the cap **34**). The container body conveys toner stored inside the container body toward the cap. The cap is provided with a through hole (e.g., the through hole **34***a*) through which a toner conveying nozzle (e.g., the toner conveying nozzle **91**) disposed in the body of the image forming apparatus is to be inserted into the container body. The through hole is disposed in a central portion of the cap. The cap has a fitted portion (e.g., the hole portion **34***b*) to be fitted with a fitting portion (e.g., the lever **97**) disposed in the body of the image forming apparatus to position the cap non-rotatably to the body of the image forming apparatus. The fitted portion is disposed in each of an upper portion and a lower portion of the cap when the cap is attached to the body of the image forming apparatus.

Second Aspect

In the toner container (e.g., the toner container **32**Y) according to the first aspect, the fitted portion (e.g., the hole portion **34***b*) is a hole portion, and the fitting portion (e.g., the lever **97**) is a lever biased by a biasing member (e.g., the torsion spring **98**) in a direction in which the fitting portion fits the hole portion. The outer diameter of the container body (e.g., the container body **33**) and the outer diameter of the cap (e.g., the cap **34**) substantially match each other.

Third Aspect

An image forming apparatus (e.g., the image forming apparatus **100**) includes: a plurality of first toner containers (e.g., the toner containers **32**Y, **32**M, and **32**C) including the toner container (e.g., the toner container **32**Y) according to the first or second aspect are arranged in the body of the image forming apparatus; and a second toner container (e.g., the toner container **32**K) adjacent to any one of the plurality of first toner containers. The second toner container includes a substantially cylindrical container body (e.g., the container body **33**) and a substantially cylindrical cap (e.g., the cap **34**). The container body conveys toner stored inside the container body toward the cap. The cap is provided with a through hole (e.g., the through hole **34***a*) through which a toner conveying nozzle (e.g., the toner conveying nozzle **91**) disposed in the body of the image forming apparatus is inserted into the container body. The through hole is disposed in a central portion of the cap. The cap has a fitted portion (e.g., the hole portion **34***b*) fitted with a fitting portion (e.g., the lever **97**) disposed in the body of the image forming apparatus to position non-rotatably to the body of the image forming apparatus.

Fourth Aspect

In the image forming apparatus (e.g., the image forming apparatus **100**) according to the third aspect, an outer diameter of the container body (e.g., the container body **33**) of the second toner container (e.g., the toner container **32**K) is larger than an outer diameter of the cap (e.g., the cap **34**) of the second toner container. An outer diameter of the container body of each of the plurality of first toner containers (e.g., the toner containers **32**Y, **32**M, and **32**C) is smaller than the outer diameter of the container body of the second toner container.

Fifth Aspect

In the image forming apparatus (e.g., the image forming apparatus **100**) according to the third or fourth aspect, the outer diameter of the cap (e.g., the cap **34**) of the second toner container (e.g., the toner container **32**K) and an outer diameter of the cap of each of the plurality of first toner containers (e.g., the toner containers **32**Y, **32**M, and **32**C) substantially match each other.

Sixth Aspect

In the image forming apparatus (e.g., the image forming apparatus **100**) according to any one of the third to fifth aspects, the fitted portion (e.g., the hole portion **34***b*) of the second toner container (e.g., the toner container **32**K) is disposed on a right side and a left side of the cap (e.g., the cap **34**) of the second toner container attached to the body of the image forming apparatus.

Seventh Aspect

In the image forming apparatus (e.g., the image forming apparatus **100**) according to any one of the third to sixth aspects, the cap (e.g., the cap **34**) of each of the plurality of first toner containers (**32**Y, **32**M, and **32**C) is provided with an identification (ID) chip on a side surface farther from the second toner container (e.g., the toner container **32**K). The identification (ID) chip can communicate with the body of the image forming apparatus. The cap of the second toner container is provided with an ID chip at a top of the cap of the second toner container. The ID chip can communicate with the body of the image forming apparatus.

Eighth Aspect

In the image forming apparatus (e.g., the image forming apparatus **100**) according to any one of the third to seventh aspects, the cap (e.g., the cap **34**) of each of the plurality of first toner containers (**32**Y, **32**M, and **32**C) is provided with a cutout portion (e.g., the cutout portion **34***c*) on a side surface closer to the second toner container (e.g., the toner container **32**K). The cutout portion is recessed as compared with a portion other than the cutout portion in the circumferential surface of the cap.

Ninth Aspect

In the image forming apparatus (e.g., the image forming apparatus **100**) according to any one of the third to eighth aspects, the cap (e.g., the cap **34**) of each of the plurality of first toner containers (**32**Y, **32**M, and **32**C) is provided with an incompatible-shape portion (e.g., the incompatible-shape portion **34***d*) that distinguishes each of the plurality of first toner containers (**32**Y, **32**M, and **32**C) from the other first toner containers of the plurality of first toner containers (**32**Y, **32**M, and **32**C).

The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention.

The invention claimed is:

1. A toner container to be attachable to and detachable from a body of an image forming apparatus, the toner container comprising:

a substantially cylindrical container body to convey toner stored inside the container body; and

a substantially cylindrical cap having a through hole through which a toner conveying nozzle disposed in the body of the image forming apparatus is to be inserted into the container body, wherein

the through hole is in a central portion of the cap,

the cap has a fitted portion to be fitted with a fitting portion disposed in the body of the image forming apparatus to position the cap non-rotatably to the body of the image forming apparatus, and

the fitted portion is disposed in each of an upper portion and a lower portion of the cap when the cap is attached to the body of the image forming apparatus.

2. The toner container according to claim 1, wherein the fitted portion is a hole portion to be fitted with a lever of the fitting portion biased by a biasing member in a direction in which the fitting portion fits the hole portion, and

an outer diameter of the container body and an outer diameter of the cap substantially match each other.

3. The toner container according to claim 1, wherein the cap of the toner container has an identification (ID) chip to communicate with the body of the image forming apparatus,

the fitted portion includes a first fitted portion on the upper portion of the cap and a second fitted portion on the lower portion of the cap, and

the ID chip is disposed between the first fitted portion and the second fitted portion when the toner container is viewed from the cap.

4. The toner container according to claim 3, wherein the ID chip is disposed in an inclined direction when the toner container is viewed from the cap.

5. The toner container according to claim 3, wherein the cap of the toner container is provided with an incompatible-shape portion for distinguishing the cap from another toner container, and

the incompatible-shape portion is disposed between the second fitted portion and the ID chip when the toner container is viewed from the cap.

6. An image forming apparatus, comprising:

a plurality of first toner containers arranged in the image forming apparatus, each first toner container of the plurality of first toner containers to be attachable to and detachable from the image forming apparatus, and each first toner container of the plurality of first toner containers including:

a container to convey toner stored inside the container; and

a cap having a through hole through which a toner conveying nozzle disposed in the image forming apparatus is inserted into the container, wherein the through hole is in a central portion of the cap, the cap has a first fitted hole fitted with a first lever disposed in the image forming apparatus and a second fitted hole with a second lever disposed in the image forming apparatus to position the cap non-rotatably to the image forming apparatus, and

the first fitted hole is disposed in an upper portion of the cap and the second fitted hole is disposed in a lower portin of the cap when the cap is attached to the image forming apparatus; and

a second toner container disposed adjacent to one first toner container of the plurality of first toner containers in the image forming apparatus, the second toner container including:

a container to convey toner stored inside the container; and

a cap having a through hole through which a toner conveying nozzle disposed in the image forming apparatus is inserted into the container, wherein the through hole is in a central portion of the cap, and the cap has a fitted hole to be fitted with a lever disposed in the image forming apparatus to position the cap non-rotatably to the image forming apparatus.

7. The image forming apparatus according to claim 6, wherein an outer diameter of the container of the second toner container is larger than an outer diameter of the cap of the second toner container, and

an outer diameter of the container of each first toner container of the plurality of first toner containers is smaller than the outer diameter of the container of the second toner container.

8. The image forming apparatus according to claim 7, wherein the outer diameter of the cap of the second toner container and the outer diameter of the cap of each first toner container of the plurality of first toner containers substantially match each other.

9. The image forming apparatus according to claim 7, wherein the fitted hole of the second toner container is disposed on a right side and a left side of the cap of the second toner container attached to the image forming apparatus.

10. The image forming apparatus according to claim 7, wherein the cap of each first toner container of the plurality of first toner containers is provided with an identification (ID) chip on a side surface farther from the second toner container, to communicate with the image forming apparatus, and

the cap of the second toner container is provided with an ID chip at a top of the cap of the second toner container, to communicate with the image forming apparatus.

11. The image forming apparatus according to claim 10, wherein the cap of each of first toner container the plurality of first toner containers is provided with a recess on a side surface closer to the second toner container, and the recess is recessed as compared with another portion of a circumferential surface of the cap.

12. The image forming apparatus according to claim 6, wherein the cap of each first toner container of the plurality of first toner containers has an incompatible-shape portion that distinguishes each first toner container of the plurality of first toner containers from other first toner containers of the plurality of first toner containers.

13. A toner container to be attachable to and detachable from of an image forming apparatus, the toner container comprising:

a container to convey toner stored inside the container; and

a cap having a through hole through which a toner conveying nozzle disposed in the image forming apparatus is to be inserted into the container, wherein

the through hole is in a central portion of the cap,

the cap bas a first fitted hole to be fitted with a first lever disposed in the image forming apparatus and a second fitted hole to be fitted with a second lever disposed in the image forming apparatus to position the cap non-rotatably to the image forming apparatus, and

the first fitted hole is disposed in an upper portion of the cap and the second fitted hole is disposed in a lower portion of the cap when the cap is attached to the image forming apparatus.

14. The toner container according to claim 13, wherein the first lever is biased by a biasing spring in a direction in which the first lever fits the first fitted hole,

the second lever is biased by a biasing spring in a direction in which the second lever fits the second fitted hole, and

an outer diameter of the container and an outer diameter of the cap match each other.

15. The toner container according to claim 13, wherein the cap of the toner container has an identification (ID) chip to communicate with the image forming apparatus, and

the ID chip is disposed between the first fitted hole and the second fitted hole when the toner container is viewed from the cap.

16. The toner container according to claim **15**, wherein the ID chip is disposed in an inclined direction when the toner container is viewed from the cap.

17. The toner container according to claim **15**, wherein

the cap is provided with an incompatible-shape portion for distinguishing the cap from another toner container, and

the incompatible-shape portion is disposed between the second fitted hole and the ID chip when the toner container is viewed from the cap.

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